Probability Distributions

Discrete & Continuous

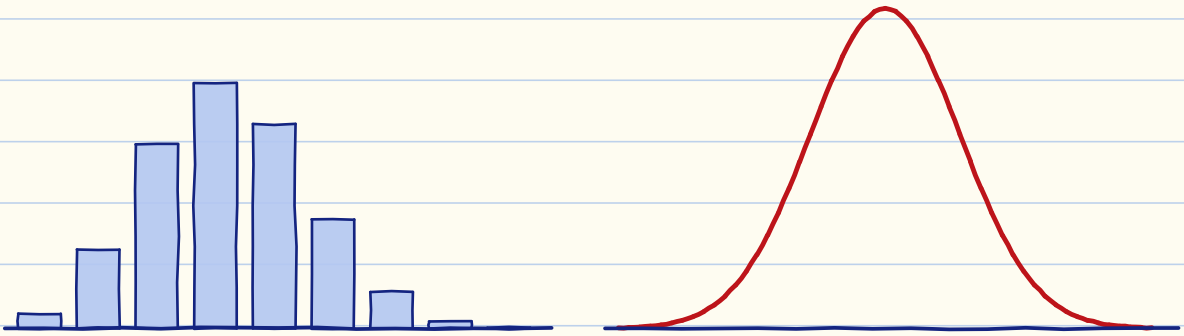
Intuition → Formulas → Excel → Solved Numericals → Practice

Part A Foundations: random variable, PMF, PDF, CDF, mean, variance

Part B Discrete: Binomial and Poisson

Part C Continuous: Uniform, Exponential and Normal

Each distribution: 8–10 solved numericals + 5 practice numericals with answer keys, Excel functions and PyTorch / TensorFlow code



discrete: count continuous: measure

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Handwritten-style study notes

Roadmap

How these notes are organised

Every distribution follows the same 5-step recipe, so revision is quick:

- **1. Intuition**: a real-life story and a picture, before any formula.
- **2. Formulas**: PMF / PDF, CDF, mean and variance, plus the logic behind them.
- **3. Excel**: exact functions, a mini worksheet, how to read the output.
- **4. Solved numericals**: step by step, each checked with an Excel formula.
- **5. Practice**: 5 numericals per distribution, with an answer key.

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Pen colours: blue = notes, red = headings and warnings, green = Excel / answers, yellow box = formula to memorise.

Tip: in Excel, always ask "single value (FALSE) or running total (TRUE)?" before typing the formula.

Foundations

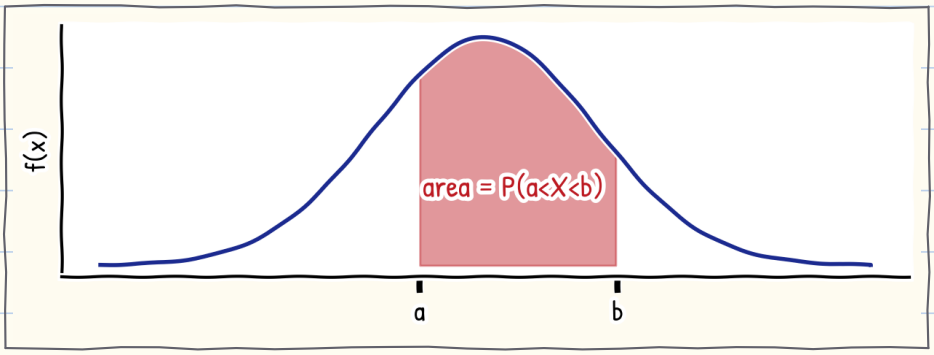
Random variables, PMF, PDF, CDF, mean and variance

1. Random variable

A random variable (RV) is a rule that converts the outcome of a random experiment into a number. Toss 3 coins and let X = number of heads. Then X can be 0, 1, 2 or 3.

2. Discrete vs continuous

	Discrete RV	Continuous RV
Values	countable: 0, 1, 2, ... (gaps)	any value in an interval (no gaps)
Comes from	COUNTING: defects, calls, heads	MEASURING: time, height, weight
Tool	PMF: P(X = x) is a real probability	PDF f(x): a density; probability = AREA
P(X = exact value)	can be greater than 0	always 0
Total	all P(x) add up to 1	total area under the curve = 1
Our examples	Binomial, Poisson	Uniform, Exponential, Normal



For a continuous RV the probability is the shaded AREA.

3. PMF, PDF and CDF

Discrete: the PMF gives each value its own probability. They are between 0 and 1 and add up to 1.

PMF

$$0 \le P(X = x) \le 1, \qquad \sum_x P(X = x) = 1$$

Continuous: probability is the area under the density curve between two points.

PDF

$$P(a \le X \le b) = \int_a^b f(x)\,dx, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$

CDF = running total of probability up to x. It works for both types.

CDF

$$F(x) = P(X \le x) = \sum_{t \le x} P(t) \quad \text{or} \quad \int_{-\infty}^{x} f(t)\,dt$$

Complement trick: $P(X > x) = 1 - F(x)$. $P(a < X \le b) = F(b) - F(a)$.

For discrete RVs: $P(X \ge k) = 1 - P(X \le k-1)$ (watch the k–1!).

4. Mean and variance

Mean = balance point of the distribution. Variance = average squared distance from the mean. SD = √variance, in the same units as X.

discrete

$$E[X] = \Sigma x\,P(x), \qquad Var(X) = E[X^2] - (E[X])^2$$

continuous

$$E[X] = \int_{-\infty}^{\infty} x\,f(x)\,dx, \qquad Var(X) = \int x^2 f(x)\,dx - \mu^2$$

Solved: mean and variance of a fair die (Excel)

	A	B	C	D
	x	P(x)	x·P(x)	x²·P(x)
2	1	0.1667	0.1667	0.1667
3	2	0.1667	0.3333	0.6667
4	3	0.1667	0.5000	1.5000
5	4	0.1667	0.6667	2.6667
6	5	0.1667	0.8333	4.1667
7	6	0.1667	1.0000	6.0000
8	Sum	1.0000	3.5000	15.1667

fx `=SUMPRODUCT(A2:A7,B2:B7)` → 3.5000 (mean)

fx `=SUMPRODUCT(A2:A7^2,B2:B7)-3.5^2` → 2.9167 (variance)

Answer: E[X] = 3.5, E[X²] = 15.1667, so Var(X) = 15.1667 − 3.5² = 35/12 = 2.9167, SD = 1.7078.

Binomial Distribution

Counting successes in n independent trials

1. Intuition

Story: flip a coin 10 times and count heads. Or test 20 bulbs and count the defective ones. Each trial has just two outcomes: success or failure.

Picture n light-switches. Each is ON with probability p, independently of the others. X = number of switches that are ON. Then X follows a Binomial(n, p) distribution, written X ~ Bin(n, p).

Check the BINS conditions before using it

- Binary: every trial has only 2 outcomes (success / failure).
- Independent: one trial does not influence another.
- Number of trials n is fixed in advance.
- Same probability p of success on every trial.

Note: "Success" only means the event you are counting. A defective bulb can be a "success" if defects are what you count.

2. Formulas and the logic behind them

One trial (Bernoulli)

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p = q$$

Build-up: one particular sequence with k successes and n−k failures has probability $p^k \cdot q^{(n-k)}$, because the trials are independent. How many such sequences exist? Choose which k of the n trials are the successes: C(n, k) ways. Multiply:

PMF

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

nCk

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

CDF

$$F(k) = P(X \le k) = \sum_{i=0}^{k} \binom{n}{i} p^i q^{n-i}$$

Mean and variance: why np and npq?

Write $X = X_1 + X_2 + \ldots + X_n$ where each X_i is a Bernoulli trial (1 if success). Each has

mean p and variance pq. Means always add; variances add because the trials are independent.

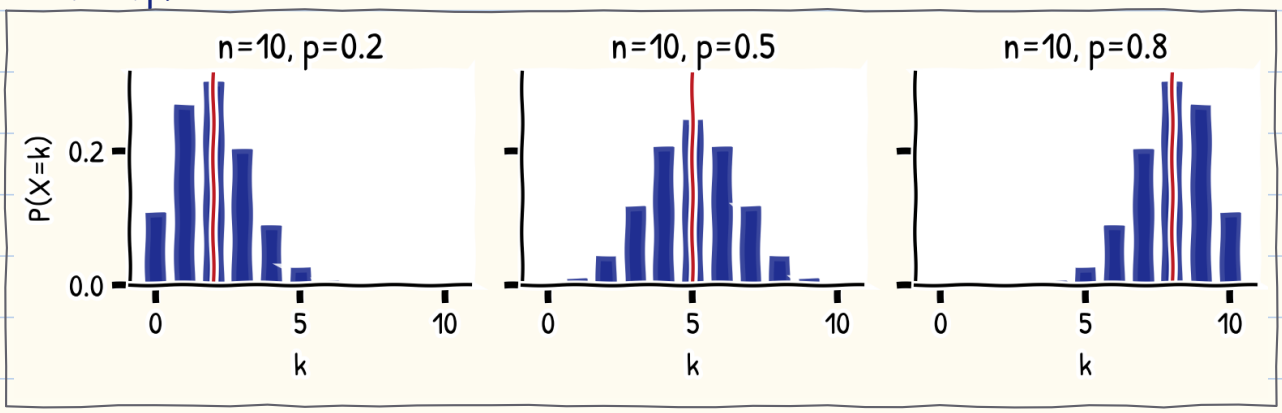
mean/var

$$E[X] = np, \qquad Var(X) = npq, \qquad \sigma = \sqrt{npq}$$

recursion

$$P(k+1) = P(k) \cdot \frac{n-k}{k+1} \cdot \frac{p}{q} \qquad \text{(starts from } P(0) = q^n\text{)}$$

The recursion lets you build the whole table by hand. The most likely value (mode) is about (n+1)p, rounded down.



p < 0.5: tail to the right. p = 0.5: symmetric. p > 0.5: tail to the left. Red line = mean np.

3. Excel calculation

Probability wanted	Excel formula
P(X = k)	=BINOM.DIST(k, n, p, FALSE)
P(X ≤ k)	=BINOM.DIST(k, n, p, TRUE)
P(X < k)	=BINOM.DIST(k-1, n, p, TRUE)
P(X ≥ k)	=1-BINOM.DIST(k-1, n, p, TRUE)
P(X > k)	=1-BINOM.DIST(k, n, p, TRUE)
P(a ≤ X ≤ b)	=BINOM.DIST.RANGE(n, p, a, b)
C(n, k)	=COMBIN(n, k)
smallest k with P(X ≤ k) ≥ α	=BINOM.INV(n, p, α)
mean, variance, SD	=n*p =n*p*(1-p) =SQRT(n*p*(1-p))

Mini worksheet: full table for n = 10, p = 0.5 (put n in F1 and p in F2).

	A	B	C	D
	k	*P(X=k)*	*P(X≤k)*	*k·P(X=k)*
2	0	0.0010	0.0010	0.0000
3	1	0.0098	0.0107	0.0098
4	2	0.0439	0.0547	0.0879
5	3	0.1172	0.1719	0.3516
6	4	0.2051	0.3770	0.8203
7	5	0.2461	0.6230	1.2305
8	6	0.2051	0.8281	1.2305
9	7	0.1172	0.9453	0.8203
10	8	0.0439	0.9893	0.3516
11	9	0.0098	0.9990	0.0879
12	10	0.0010	1.0000	0.0098

fx `B2: =BINOM.DIST(A2,$F$1,$F$2,FALSE)`

fx `C2: =BINOM.DIST(A2,$F$1,$F$2,TRUE)`

fx `D2: =A2*B2` then `=SUM(D2:D12)` → 5.0000 (= np)

Excel trick: Simulation: =BINOM.INV(10,0.5,RAND()) gives a fresh random Binomial value every time you press F9.

4. Solved numericals

Example 1. A fair coin is tossed 10 times. Find (a) P(exactly 4 heads), (b) P(at least 8 heads).

Given: n = 10, p = q = 0.5, X ~ Bin(10, 0.5).

(a)

$$P(X=4) = \binom{10}{4}(0.5)^4(0.5)^6 = 210 \times \frac{1}{1024} = 0.2051$$

fx `=BINOM.DIST(4,10,0.5,FALSE)` → 0.2051

(b) "At least 8" means 8, 9 or 10 heads.

$$P(X \geq 8) = \frac{45+10+1}{1024} = \frac{56}{1024} = 0.0547$$

fx `=1-BINOM.DIST(7,10,0.5,TRUE)` → 0.0547

Answer: (a) 0.2051 (b) 0.0547

Example 2. 5% of bulbs from a factory are defective. A sample of 20 is drawn. Find (a) P(none defective), (b) P(at most 2 defective), (c) P(at least 1 defective), (d) mean and SD.

Given: n = 20, p = 0.05, q = 0.95. Defective = "success".

$$\text{(a) } P(X = 0) = (0.95)^{20} = 0.3585$$

$$\text{(b) } P(X \le 2) = 0.3585 + 0.3774 + 0.1887 = 0.9245$$

`fx` `=BINOM.DIST(2,20,0.05,TRUE)` → 0.9245

$$\text{(c) } P(X \ge 1) = 1 - P(X = 0) = 1 - 0.3585 = 0.6415$$

$$\text{(d) } \mu = np = 1, \quad \sigma^2 = npq = 0.95, \quad \sigma = 0.9747$$

Answer: (a) 0.3585 (b) 0.9245 (c) 0.6415 (d) mean 1, SD 0.9747

Example 3. A test has 12 multiple-choice questions with 4 options each. A student guesses every answer. Pass mark = 6 correct. Find P(pass) and the expected score with its SD.

Given: n = 12, p = 1/4 (one correct option out of four), X ~ Bin(12, 0.25).

$$P(X \ge 6) = 1 - P(X \le 5) = 1 - 0.9456 = 0.0544$$

`fx` `=1-BINOM.DIST(5,12,0.25,TRUE)` → 0.0544

$$\mu = np = 3, \qquad \sigma = \sqrt{12 \times 0.25 \times 0.75} = 1.5000$$

Answer: P(pass by guessing) = 0.0544 (about 5%). Expected score 3, SD 1.5000.

Careful! P(X ≥ 6) is 1 − P(X ≤ 5), not 1 − P(X ≤ 6). In Excel the TRUE function stops at the number you give it.

4b. More solved numericals: going deeper

Example 4 (build the PMF by recursion). X ~ Bin(5, 0.4). Build the whole table by hand using P(k+1) = P(k) × (n−k)/(k+1) × p/q.

Start: P(0) = 0.6^5 = 0.0778. Ratio factor p/q = 0.4/0.6 = 0.6667.

P(1) = 0.0778 × (5/1) × 0.6667 = 0.2592

P(2) = 0.2592 × (4/2) × 0.6667 = 0.3456

P(3) = 0.3456 × (3/3) × 0.6667 = 0.2304

P(4) = 0.2304 × (2/4) × 0.6667 = 0.0768

P(5) = 0.0768 × (1/5) × 0.6667 = 0.0102

`fx` `=BINOM.DIST(3,5,0.4,FALSE)` → 0.2304 (check P(3))

Answer: Values 0.0778, 0.2592, 0.3456, 0.2304, 0.0768, 0.0102; they add to 1.0000. Mode = 2 (largest value 0.3456).

Example 5 (how many trials are needed?). A sales call succeeds with probability 0.10. How many independent calls are needed so that P(at least one sale) ≥ 0.95?

$$P(X \geq 1) = 1 - (0.9)^n \geq 0.95 \Rightarrow (0.9)^n \leq 0.05 \Rightarrow n \geq \frac{\ln 0.05}{\ln 0.9} = 28.43$$

`fx` `=1-BINOM.DIST(0,28,0.1,TRUE)` → 0.9477 (n = 28: not enough)

`fx` `=1-BINOM.DIST(0,29,0.1,TRUE)` → 0.9529 (n = 29: enough)

Answer: n = 29 calls. Logic: "at least one" is easiest through the complement, 1 – q^n.

Example 6 (most likely value). X ~ Bin(15, 0.3). Find the mode and confirm it by comparing neighbours.

Mode = floor((n+1)p) = floor(16 × 0.3) = floor(4.8) = 4.

Check: P(3) = 0.1700, P(4) = 0.2186, P(5) = 0.2061. The peak is at 4, while the mean np = 4.5 lies between 4 and 5.

Example 7 (money at risk). A lender has 8 independent loans, each defaulting with probability 0.10. Each default costs ₹2 lakh. Find the expected loss, its SD, and P(loss ≥ ₹6 lakh).

Let X = number of defaults ~ Bin(8, 0.1). Loss L = 2X. Scaling rules: E[cX] = c·E[X] and Var(cX) = c²·Var(X).

$$E[L] = 2 \times 0.8 = 1.6 \text{ lakh}, \qquad \sigma_L = 2\sqrt{8 \times 0.1 \times 0.9} = 1.6971$$

Loss ≥ 6 lakh means at least 3 defaults.

`fx` `=1-BINOM.DIST(2,8,0.1,TRUE)` → 0.0381

Answer: Expected loss ₹1.6 lakh, SD ₹1.697 lakh, P(loss ≥ ₹6 lakh) = 0.0381.

Example 8 (within one SD). $X \sim \text{Bin}(20, 0.5)$. Find $P(\mu - \sigma \le X \le \mu + \sigma)$ and compare with the normal 68% rule.

$\mu = 10$, $\sigma = \sqrt{5} = 2.236$, so the integers from 7.76 to 12.24, that is 8 to 12.

`fx` `=BINOM.DIST(12,20,0.5,TRUE)-BINOM.DIST(7,20,0.5,TRUE)` → 0.7368

Answer: 0.7368, a little above 68% because the bars are discrete (whole numbers only).

5. Practice: 5 numericals

- A fair die is rolled 8 times. Find P(exactly two sixes).
- 15% of software builds fail. In 10 builds find P(at most 1 fails).
- 60% of customers prefer UPI. Among 15 customers find P(at least 10 prefer UPI).
- A player makes 80% of free throws. In 6 attempts find P(at least 5 made).
- $X \sim \text{Bin}(40, 0.3)$. Find mean, variance, SD and $P(X = 12)$.

Answer key (try on your own first, then check!)

1. =BINOM.DIST(2,8,1/6,FALSE) → 0.2605
2. =BINOM.DIST(1,10,0.15,TRUE) → 0.5443
3. =1-BINOM.DIST(9,15,0.6,TRUE) → 0.4032
4. =1-BINOM.DIST(4,6,0.8,TRUE) → 0.6554
5. mean 12, variance 8.4, SD 2.8983; =BINOM.DIST(12,40,0.3,FALSE) → 0.1366

Code corner: Binomial

SciPy

```
from scipy.stats import binom
n, p = 10, 0.5
binom.pmf(4, n, p)        # P(X=4)  = 0.2051
binom.cdf(4, n, p)        # P(X<=4)
1 - binom.cdf(7, n, p)    # P(X>=8) = 0.0547
binom.mean(n, p), binom.var(n, p)
```

PyTorch

```
import torch
from torch.distributions import Binomial
d = Binomial(total_count=10, probs=torch.tensor(0.5))
d.log_prob(torch.tensor(4.)).exp()            # P(X=4)
d.log_prob(torch.arange(0., 5.)).exp().sum()  # P(X<=4)
d.mean, d.variance
```

TensorFlow Probability

```
import tensorflow_probability as tfp

d = tfp.distributions.Binomial(total_count=10., probs=0.5)

d.prob(4.)     # P(X=4)

d.cdf(4.)      # P(X<=4)

d.mean(), d.variance()
```

Poisson Distribution

Counting events in a fixed interval of time or space

1. Intuition

Story: calls arrive at a help-desk at an average of 3 per minute. In any given minute you might see 0, 1, 2, 5, even 9 calls. The Poisson distribution gives the probability of each count.

Picture raindrops landing on one paving tile. Drops fall independently and at a steady average rate. Count the drops on the tile in one minute: that count is Poisson(λ), where λ is the average number per interval.

Typical situations

- calls per minute, emails per hour, customers per 10 minutes
- typing errors per page, defects per metre of cable, bugs per 1000 lines
- accidents per week, goals per match, server errors per day

Conditions

- events occur independently of each other
- the average rate is constant (does not change over time)
- two events cannot happen at exactly the same instant
- λ is proportional to the length of the interval: rate r per unit, interval t gives $\lambda = r \cdot t$

2. Formulas and the logic behind them

Where does it come from? Cut the interval into n tiny slices. In each slice an event happens with small probability $p = \lambda/n$. The count is Binomial(n, λ/n). Let $n \to \infty$ and the binomial turns into the Poisson:

$$\binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k} \longrightarrow \frac{e^{-\lambda}\lambda^k}{k!}$$

PMF

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

CDF

$$F(k) = P(X \le k) = e^{-\lambda}\sum_{i=0}^{k}\frac{\lambda^i}{i!}$$

meanvar

$$E[X] = \lambda, \qquad Var(X) = \lambda, \qquad \sigma = \sqrt{\lambda}$$

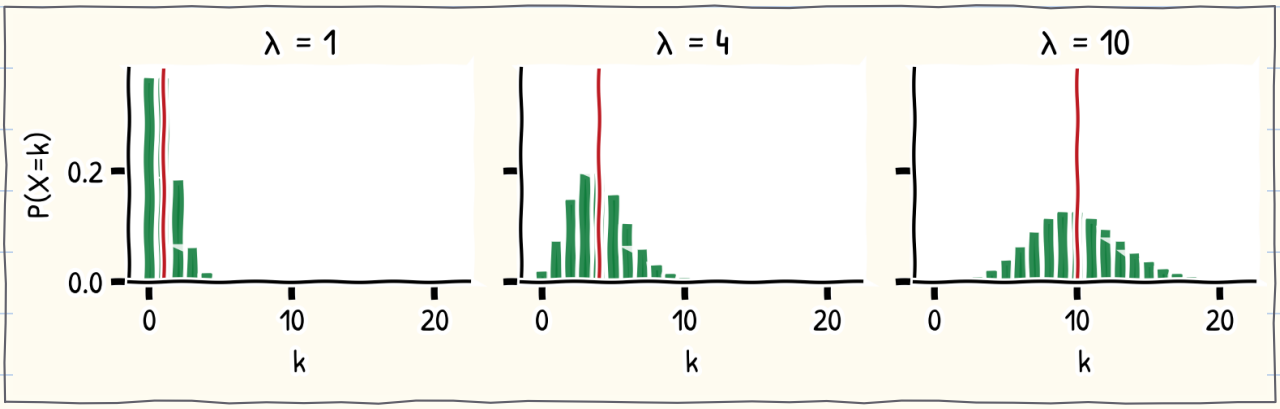
Why mean = λ? Sum k·P(k): the k = 0 term vanishes, k/k! = 1/(k−1)!, pull out λ and e^(−λ), and what remains is the series of e^λ:

$$E[X] = \sum_{k=1}^{\infty} k\frac{e^{-\lambda}\lambda^k}{k!} = \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda}e^{\lambda} = \lambda$$

recursion

$$P(k+1) = P(k)\cdot\frac{\lambda}{k+1}, \qquad P(0) = e^{-\lambda}$$

- Mean = variance = λ. Real data with variance much larger than the mean is not Poisson.
- Sum of independent Poissons is Poisson: Poisson(λ_1) + Poisson(λ_2) = Poisson(λ_1 + λ_2).
- Mode = floor(λ). If λ is a whole number, both λ−1 and λ are modes.



Small λ: very skewed. Large λ: nearly symmetric, bell-like.

3. Excel calculation

Probability wanted	Excel formula
P(X = k)	=POISSON.DIST(k, λ, FALSE)
P(X ≤ k)	=POISSON.DIST(k, λ, TRUE)
P(X < k)	=POISSON.DIST(k-1, λ, TRUE)
P(X ≥ k)	=1-POISSON.DIST(k-1, λ, TRUE)
P(a ≤ X ≤ b)	=POISSON.DIST(b,λ,TRUE)-POISSON.DIST(a-1,λ,TRUE)
manual check	=EXP(-λ)*λ^k/FACT(k)
new λ for a longer interval	=rate*t

Remember: Excel has no POISSON.INV. To find a percentile, build the cumulative column and look for the first value that reaches α.

Mini worksheet: $\lambda = 4$ (put λ in F1).

	A	B	C
	k	$P(X=k)$	$P(X\le k)$
2	0	0.0183	0.0183
3	1	0.0733	0.0916
4	2	0.1465	0.2381
5	3	0.1954	0.4335
6	4	0.1954	0.6288
7	5	0.1563	0.7851
8	6	0.1042	0.8893
9	7	0.0595	0.9489
10	8	0.0298	0.9786
11	9	0.0132	0.9919
12	10	0.0053	0.9972

fx `B2: =POISSON.DIST(A2,$F$1,FALSE)`

fx `C2: =POISSON.DIST(A2,$F$1,TRUE)`

4. Solved numericals

Example 1. A help-desk receives 3 calls per minute on average. Find the probability of (a) exactly 5 calls in a minute, (b) at most 2 calls, (c) at least 1 call.

Given: $\lambda = 3$, $X \sim \text{Poisson}(3)$.

$$\text{(a) } P(X=5) = \frac{e^{-3}3^5}{5!} = \frac{0.049787 \times 243}{120} = 0.1008$$

fx `=POISSON.DIST(5,3,FALSE)` → 0.1008

$$\text{(b) } P(X \le 2) = 0.0498 + 0.1494 + 0.2240 = 0.4232$$

fx `=POISSON.DIST(2,3,TRUE)` → 0.4232

$$\text{(c) } P(X \ge 1) = 1 - P(0) = 1 - e^{-3} = 0.9502$$

Answer: (a) 0.1008 (b) 0.4232 (c) 0.9502

Example 2. A cable has on average 2 defects per 100 m. Find (a) P(no defect in 100 m), (b) P(at least 3 defects in 100 m), (c) P(no defect in 250 m).

$$\text{(a) } \lambda = 2: \quad P(X = 0) = e^{-2} = 0.1353$$

$$\text{(b) } P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.6767 = 0.3233$$

`fx` `=1-POISSON.DIST(2,2,TRUE)` → 0.3233

(c) Longer interval, so scale the rate: $\lambda = 2 \times 2.5 = 5$.

$$P(X = 0) = e^{-5} = 0.0067$$

Answer: (a) 0.1353 (b) 0.3233 (c) 0.0067

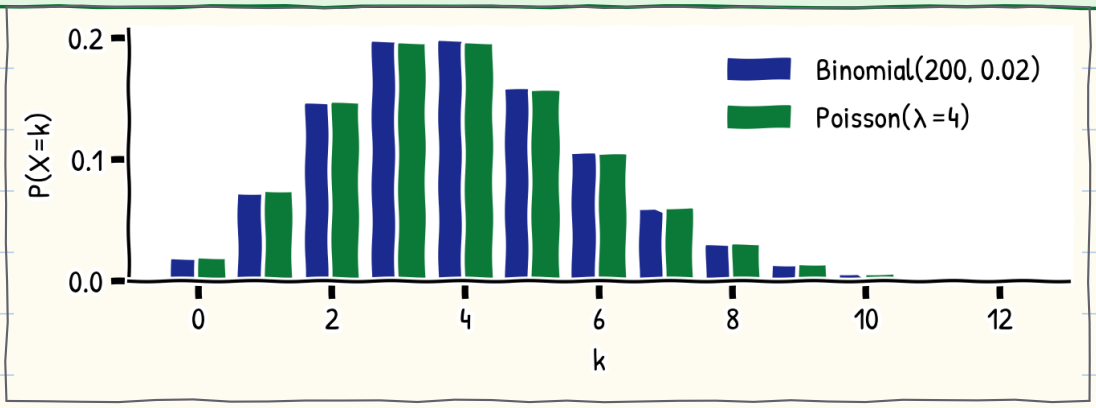
Example 3 (Poisson approximation). 2% of items are faulty. In a batch of 200 find P(exactly 3 faulty) exactly and by the Poisson approximation.

Exact: $n = 200$, $p = 0.02$. Approximate: $\lambda = np = 4$. Rule of thumb: n large (≥ 20) and p small (≤ 0.05).

`fx` `=BINOM.DIST(3,200,0.02,FALSE)` → 0.1963 (exact)

`fx` `=POISSON.DIST(3,4,FALSE)` → 0.1954 (approx)

Answer: Exact 0.1963 vs Poisson 0.1954. The approximation is close and much easier by hand.



Blue bars (binomial) and green bars (Poisson) almost coincide.

4b. More solved numericals: going deeper

Example 4 (find λ from data). A shop sees no customer in a given minute 8.21% of the time. Assuming Poisson arrivals, find λ and P(at least 3 customers in a minute).

$$P(X=0) = e^{-\lambda} = 0.0821 \Rightarrow \lambda = -\ln(0.0821) = 2.4998 \approx 2.5$$

fx `=1-POISSON.DIST(2,2.5,TRUE)` → 0.4562

Answer: $\lambda \approx 2.5$ per minute, $P(X \geq 3) = 0.4562$.

Example 5 (adding Poissons). Machine A makes 1.2 faults per shift and machine B makes 2.3, independently. Find P(total faults ≤ 2).

Sum of independent Poissons is Poisson with $\lambda = 1.2 + 2.3 = 3.5$.

fx `=POISSON.DIST(2,1.2+2.3,TRUE)` → 0.3208

Answer: 0.3208. Never multiply or average the probabilities; add the rates first.

Example 6 (tied modes). $X \sim$ Poisson(4). Use the recursion $P(k+1) = P(k) \times \lambda/(k+1)$ to find the mode.

$P(3) = 0.1954$. Next: $P(4) = P(3) \times 4/4 = 0.1954$. The ratio is exactly 1, so $P(3) = P(4)$.

Answer: Two modes: 3 and 4. This always happens when λ is a whole number (modes $\lambda - 1$ and λ).

Example 7 (conditional probability). Calls follow Poisson(2) per minute. Given that at least one call arrived, find P(exactly one call).

$$P(X=1 \mid X \geq 1) = \frac{P(X=1)}{1-P(X=0)} = \frac{0.2707}{0.8647} = 0.3130$$

Example 8 (capacity planning). A desk handles at most 6 calls a minute; $\lambda = 3$. Find P(overload). What capacity c keeps P(overload) $\leq$ 1%?

fx `=1-POISSON.DIST(6,3,TRUE)` → 0.0335 (c = 6)

Try larger c: $P(X > 7) = 0.0119$, $P(X > 8) = 0.0038$, $P(X > 9) = 0.0011$.

Answer: Overload probability 0.0335 now. Capacity c = 8 brings it to 0.0038 ($\leq$ 1%).

Example 9 (normal approximation to Poisson). $X \sim$ Poisson(25). Find $P(X \leq 30)$ exactly and with N(25, 25).

$$z = \frac{30.5-25}{\sqrt{25}} = 1.1 \Rightarrow \Phi(1.1) = 0.8643$$

fx `=POISSON.DIST(30,25,TRUE)` → 0.8633 (exact)

Answer: Approximation 0.8643 vs exact 0.8633.

5. Practice: 5 numericals

- A server logs 2 errors per day on average. Find P(no error tomorrow).
- Buses arrive at 6 per hour. Find P(8 or more buses in an hour).
- A page has 1.5 typing errors on average. Find (a) P(at most 1 error on a page), (b) P(exactly 3 errors in 2 pages).
- Accidents occur at 0.5 per week. Find P(at least 1 accident in 4 weeks).
- n = 500, p = 0.01. Use the Poisson approximation for $P(X \le 3)$ and compare with the exact value.

Answer key (try on your own first, then check!)

1. =POISSON.DIST(0,2,FALSE) → 0.1353
2. =1-POISSON.DIST(7,6,TRUE) → 0.2560
3a. =POISSON.DIST(1,1.5,TRUE) → 0.5578 3b. =POISSON.DIST(3,3,FALSE) → 0.2240
4. $\lambda = 2$: =1-POISSON.DIST(0,2,TRUE) → 0.8647
5. $\lambda = 5$: approx 0.2650; exact =BINOM.DIST(3,500,0.01,TRUE) → 0.2636

Code corner: Poisson

SciPy

```
from scipy.stats import poisson
lam = 4
poisson.pmf(3, lam)        # P(X=3)
poisson.cdf(2, lam)        # P(X<=2)
1 - poisson.cdf(0, lam)    # P(X>=1)
poisson.mean(lam), poisson.var(lam)
```

PyTorch

```
import torch
from torch.distributions import Poisson
d = Poisson(rate=torch.tensor(4.))
d.log_prob(torch.tensor(3.)).exp()            # P(X=3)
d.log_prob(torch.arange(0., 3.)).exp().sum()  # P(X<=2)
d.mean, d.variance
```

TensorFlow Probability

```
import tensorflow_probability as tfp
d = tfp.distributions.Poisson(rate=4.)
d.prob(3.)     # P(X=3)
d.cdf(2.)      # P(X<=2)
d.mean(), d.variance()
```

Binomial vs Poisson at a glance

	Binomial	Poisson
Counts	successes in n fixed trials	events in a fixed interval
Possible values	0, 1, ..., n (upper limit)	0, 1, 2, ... (no upper limit)
Parameters	n and p	λ only
Mean / variance	np / npq (variance < mean)	λ / λ (equal)
Link	n large, p small: Bin(n, p) ≈ Poisson(np)	

Continuous Probability

From counting to measuring

1. Intuition

Between 1.0 and 1.1 minutes there are infinitely many possible times (1.05, 1.0501, ...). We cannot list a probability for each one. So we spread the probability along the number line like paint. The thickness of the paint at x is the density f(x).

- A tall part of the curve = values there are more likely (per unit length).
- Probability is the area under the curve over an interval, not the height.
- A single point has zero width, so zero area: $P(X = x) = 0$.
- Consequence: $P(a < X < b) = P(a \le X \le b)$. Strict or not, nobody cares.

Careful! f(x) can be bigger than 1 (a very narrow, tall curve). That is fine, because it is a density. Only the AREA must be at most 1.

2. Rules to remember

PDF

$$f(x) \ge 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$

CDF

$$F(x) = \int_{-\infty}^{x} f(t)\,dt, \qquad f(x) = \frac{dF(x)}{dx}$$

$$P(a < X < b) = F(b) - F(a), \qquad P(X > a) = 1 - F(a)$$

3. Which continuous distribution?

Distribution	Use when	Shape
Uniform $U(a, b)$	every value in [a, b] is equally likely (random arrival within a window)	flat rectangle
Exponential $Exp(\lambda)$	waiting time until the next event; memoryless	starts high, decays
Normal $N(\mu, \sigma^2)$	natural variation, measurement error, sums and averages	symmetric bell

Plan: Excel summary: Uniform has no special function (use simple formulas), Exponential uses EXPON.DIST, Normal uses NORM.DIST / NORM.S.DIST / NORM.INV.

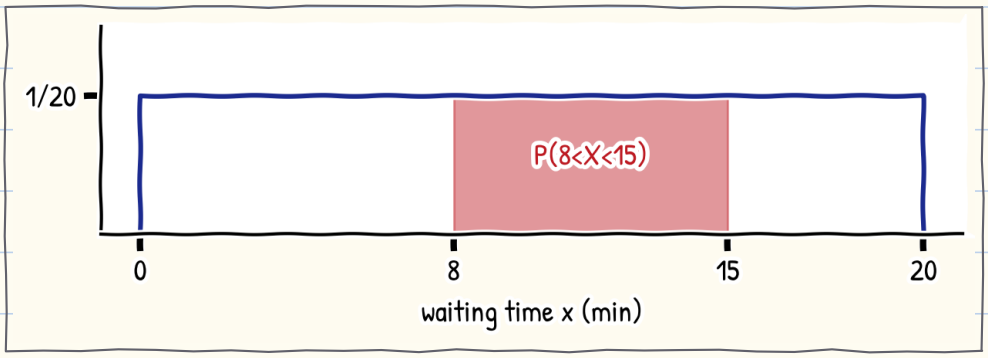
Uniform Distribution

Every value between a and b is equally likely

1. Intuition

A bus is due at some random moment in the next 20 minutes, and every moment is equally likely. A random-number generator, or the rounding error in a measurement, behaves the same way. The density is a flat rectangle over [a, b].

Area must be 1, and the width is (b − a), so the height has to be 1/(b − a). That is the whole distribution.



U(0, 20): height 1/20 = 0.05. Shaded area = 7 × 0.05 = 0.35.

2. Formulas and the logic behind them

PDF

$$f(x) = \frac{1}{b-a} \quad (a \le x \le b), \quad 0 \text{ otherwise}$$

CDF

$$F(x) = \frac{x-a}{b-a} \quad (a \le x \le b), \quad 0\ (x < a), \quad 1\ (x > b)$$

$$P(c \le X \le d) = \frac{d-c}{b-a} \qquad \text{(length ratio)}$$

mean/var

$$E[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}, \qquad \sigma = \frac{b-a}{\sqrt{12}}$$

Why the midpoint? By symmetry the balance point is the centre. Check:

$$E[X] = \int_a^b \frac{x}{b-a}\,dx = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$$

Variance: $E[X^2] = (a^2 + ab + b^2)/3$, so $\mathrm{Var} = (a^2 + ab + b^2)/3 - ((a + b)/2)^2 = (b - a)^2/12$.

median $\text{median} = \frac{a+b}{2}, \quad p\text{-th percentile: } x_p = a + p(b-a)$

3. Excel calculation

There is no UNIFORM.DIST in Excel. Type the formulas directly (a in B1, b in B2):

Quantity	Excel formula
density f(x)	=1/(b-a)
P(X ≤ x)	=(x-a)/(b-a)
P(c ≤ X ≤ d)	=(d-c)/(b-a)
P(X > x)	=(b-x)/(b-a)
mean, variance, SD	=(a+b)/2 =(b-a)^2/12 =(b-a)/SQRT(12)
percentile p	=a+p*(b-a)
simulation	=a+(b-a)*RAND()

Worksheet for U(0, 20):

	A	B	C
	x	*f(x)*	*F(x)*
2	0	0.0500	0.0000
3	5	0.0500	0.2500
4	8	0.0500	0.4000
5	10	0.0500	0.5000
6	15	0.0500	0.7500
7	20	0.0500	1.0000

fx `B2: =1/($F$2-$F$1)`

fx `C2: =(A2-$F$1)/($F$2-$F$1)` → F1 = 0 (a), F2 = 20 (b)

4. Solved numericals

Example 1. A bus arrives uniformly in the next 20 minutes, $X \sim U(0, 20)$. Find $P(X < 5)$, $P(8 < X < 15)$, the mean and the SD.

$$P(X<5) = \frac{5}{20} = 0.2500, \qquad P(8<X<15) = \frac{15-8}{20} = 0.3500$$

$$\mu = \frac{0+20}{2} = 10, \qquad \sigma = \frac{20}{\sqrt{12}} = 5.7735$$

fx `=(15-8)/(20-0)` → 0.3500

Answer: 0.2500, 0.3500, mean 10 min, SD 5.7735 min.

Example 2. Delivery time is uniform between 30 and 60 minutes. Find $P(X > 50)$, $P(35 \le X \le 45)$, the median and the variance.

$$P(X>50) = \frac{60-50}{30} = 0.3333, \quad P(35 \le X \le 45) = \frac{10}{30} = 0.3333$$

$$\text{median} = \frac{30+60}{2} = 45, \qquad \text{Var} = \frac{30^2}{12} = 75$$

Answer: 0.3333, 0.3333, median 45 min, variance 75 (SD 8.6603).

Example 3. $X \sim U(2, 12)$. Find k such that $P(X > k) = 0.10$.

$$\frac{12-k}{10} = 0.10 \Rightarrow k = 12 - 1 = 11 \qquad (\text{same as } x_{0.9} = 2 + 0.9 \times 10)$$

fx	=2+0.9*(12-2)	→ 11

Example 4 (find a and b). A uniform RV has mean 15 and variance 75. Find a and b.

$$\frac{a+b}{2} = 15 \Rightarrow a + b = 30; \qquad \frac{(b-a)^2}{12} = 75 \Rightarrow b - a = 30$$

Answer: Adding: 2b = 60, so b = 30 and a = 0. The distribution is U(0, 30).

4b. More solved numericals: going deeper

Example 5 (conditional probability). $X \sim U(0, 20)$. Given that $X > 8$, find $P(X < 15)$.

Once we know $X > 8$, the possible range shrinks to [8, 20] and X is still uniform there.

$$P(X<15 \mid X>8) = \frac{P(8<X<15)}{P(X>8)} = \frac{7/20}{12/20} = \frac{7}{12} = 0.5833$$

Example 6 (linear change of units). $X \sim U(0, 10)$ and $Y = 2X + 5$. Find the distribution, mean and variance of Y.

Stretching by 2 and shifting by 5 maps [0, 10] to [5, 25], so $Y \sim U(5, 25)$.

$$E[Y] = 2(5) + 5 = 15 = \frac{5+25}{2}, \qquad \text{Var}(Y) = 2^2 \times \frac{10^2}{12} = 33.3333 = \frac{20^2}{12}$$

Answer: Rules: $E[aX + b] = aE[X] + b$ and $\text{Var}(aX + b) = a^2\text{Var}(X)$. The shift b never changes the spread.

Example 7 (inventory decision). Weekly demand $D \sim U(100, 200)$. You stock 150 units. Find P(shortage) and the expected shortage $E[\max(D - 150, 0)]$.

P(shortage) = $P(D > 150) = 50/100 = 0.5$. For the expected shortage integrate the shortfall over D from 150 to 200:

$$E[\text{shortage}] = \int_{150}^{200} (d - 150)\frac{1}{100}\,dd = \frac{50^2}{2 \times 100} = 12.5$$

Answer: Shortage happens half the time, and averages 12.5 units per week. Raising stock to 175 cuts it to $25^2/200 = 3.125$.

Example 8 (probability of a sub-interval inside a window). You arrive at a station at a random time during a 30-minute window when the next train is equally likely to come any time. Find P(the train comes in the first 10 minutes or the last 5 minutes). Two disjoint intervals, so add: 10/30 + 5/30.

Answer: 0.5000 (lengths add: 15 minutes out of 30).

5. Practice: 5 numericals

- $X \sim U(0, 10)$. Find $P(X > 7)$.
- $X \sim U(5, 25)$. Find the mean, variance and SD.
- A train comes every 15 minutes and you arrive at a random time. Find P(wait between 3 and 8 minutes) and the average wait.
- $X \sim U(20, 80)$. Find $P(30 < X < 50)$.
- $X \sim U(100, 200)$. Find the 75th percentile and $P(X < 130)$.

Answer key (try on your own first, then check!)

1. =(10-7)/(10-0) → 0.3000
2. mean 15, variance 33.3333, SD 5.7735
3. =(8-3)/15 → 0.3333; average wait 7.5 min
4. =(50-30)/(80-20) → 0.3333
5. =100+0.75*100 → 175; =(130-100)/100 → 0.3000

Code corner: Uniform

SciPy

```
from scipy.stats import uniform
d = uniform(loc=0, scale=20)    # U(0, 20): loc=a, scale=b-a
d.cdf(5)                # P(X<5)  = 0.25
d.cdf(15) - d.cdf(8)    # P(8<X<15) = 0.35
d.mean(), d.std()
```

PyTorch

```
import torch
from torch.distributions import Uniform
d = Uniform(0., 20.)
d.cdf(torch.tensor(5.))
d.cdf(torch.tensor(15.)) - d.cdf(torch.tensor(8.))
d.mean, d.stddev
```

TensorFlow Probability

```
import tensorflow_probability as tfp
d = tfp.distributions.Uniform(low=0., high=20.)
d.cdf(5.)
d.cdf(15.) - d.cdf(8.)
d.mean(), d.stddev()
```

Exponential Distribution

Waiting time until the next event

1. Intuition

Calls arrive at a rate of λ per minute (a Poisson process). How long until the next call? That waiting time follows the exponential distribution. Short waits are most likely, and long waits get rarer and rarer, so the curve starts high and decays.

Link with Poisson (the key idea)

"Waiting longer than t" means "zero events in the first t units". The number of events in t units is Poisson(λt), so:

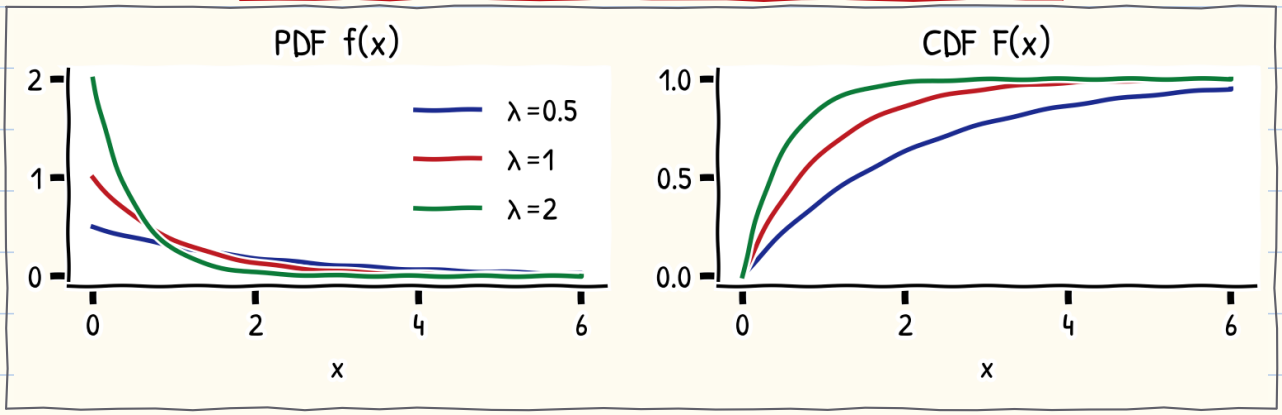
survival

$$P(X > t) = P(\text{no event in } t) = \frac{e^{-\lambda t}(\lambda t)^0}{0!} = e^{-\lambda t}$$

Memoryless property

A bus with no timetable: if you have already waited 10 minutes, the remaining wait has the same distribution as when you arrived. The past does not matter.

$$P(X > s+t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$



Bigger rate λ: waits are shorter, the curve falls faster.

2. Formulas

PDF/CDF

$$f(x) = \lambda e^{-\lambda x},\ x \geq 0 \qquad F(x) = 1 - e^{-\lambda x} \qquad P(X > x) = e^{-\lambda x}$$

mean/var

$$E[X] = \frac{1}{\lambda}, \qquad \text{Var}(X) = \frac{1}{\lambda^2}, \qquad \sigma = \frac{1}{\lambda}$$

percentiles

$$\text{median} = \frac{\ln 2}{\lambda}, \qquad x_p = \frac{-\ln(1-p)}{\lambda}$$

Why 1/λ? Integration by parts on ∫ x λ e^(−λx) dx gives 1/λ, and E[X²] = 2/λ², so Var = 2/λ² − 1/λ² = 1/λ². Check with sense: if 3 calls come per minute, the average gap is 1/3 minute.

Careful! Two ways to describe the same thing: rate λ (events per unit time) or mean θ = 1/λ (time per event). Excel wants the RATE λ.

3. Excel calculation

Quantity	Excel formula
density f(x)	=EXPON.DIST(x, λ, FALSE)
P(X ≤ x)	=EXPON.DIST(x, λ, TRUE)
P(X > x)	=1-EXPON.DIST(x, λ, TRUE) or =EXP(-λ*x)
P(a < X < b)	=EXPON.DIST(b,λ,TRUE)-EXPON.DIST(a,λ,TRUE)
mean / SD	=1/λ
median	=LN(2)/λ
percentile p	=-LN(1-p)/λ
simulation	=-LN(RAND())/λ

Worksheet for λ = 0.5 (mean 2 minutes):

	A	B	C	D
	x	f(x)	F(x)	P(X>x)
2	0	0.5000	0.0000	1.0000
3	1	0.3033	0.3935	0.6065
4	2	0.1839	0.6321	0.3679
5	3	0.1116	0.7769	0.2231
6	4	0.0677	0.8647	0.1353
7	5	0.0410	0.9179	0.0821
8	6	0.0249	0.9502	0.0498

fx B2: =EXPON.DIST(A2,0.5,FALSE)

fx C2: =EXPON.DIST(A2,0.5,TRUE)

fx D2: =1-C2

4. Solved numericals

Example 1. Customers arrive with an average gap of 2 minutes. Find P(wait > 3 min) and P(1 < wait < 4).

Mean 2 min, so $\lambda = 1/2 = 0.5$ per minute.

$$P(X > 3) = e^{-0.5 \times 3} = e^{-1.5} = 0.2231$$

$$P(1 < X < 4) = e^{-0.5} - e^{-2} = 0.6065 - 0.1353 = 0.4712$$

`fx` `=EXPON.DIST(4,0.5,TRUE)-EXPON.DIST(1,0.5,TRUE)` → 0.4712

Answer: 0.2231 and 0.4712

Example 2. A bulb has mean life 1000 hours ($\lambda = 0.001$). Find (a) P(lasts more than 1500 h), (b) P(fails within 500 h), (c) given it has already survived 800 h, P(it lasts 500 h more).

$$\text{(a) } e^{-1.5} = 0.2231 \qquad \text{(b) } 1 - e^{-0.5} = 0.3935$$

(c) Memoryless! The 800 hours already survived are forgotten:

$$P(X > 1300 \mid X > 800) = P(X > 500) = e^{-0.5} = 0.6065$$

Answer: (a) 0.2231 (b) 0.3935 (c) 0.6065

Example 3. Failures arrive at $\lambda = 0.2$ per hour. Find the median wait, the 90th percentile and P(X > mean).

$$\text{median} = \frac{\ln 2}{0.2} = 3.4657, \qquad x_{0.9} = \frac{-\ln 0.1}{0.2} = 11.5129$$

`fx` `=-LN(1-0.9)/0.2` → 11.5129

P(X > mean) = e^(−λ·(1/λ)) = e^(−1) = 0.3679. Only 36.8% of waits exceed the average, because the curve is skewed to the right.

4b. More solved numericals: going deeper

Example 4 (find λ from a probability). A call lasts longer than 10 minutes with probability 0.30. Assuming exponential durations, find λ, the mean duration and P(a call exceeds 20 minutes).

$$e^{-10\lambda} = 0.30 \Rightarrow \lambda = \frac{-\ln 0.30}{10} = 0.1204, \quad \text{mean} = \frac{1}{\lambda} = 8.31$$

`fx` `=-LN(0.3)/10` → 0.1204

Answer: $\lambda = 0.1204$ per minute, mean 8.31 min. $P(X > 20) = 0.30^2 = 0.0900$ (surviving two 10-minute blocks, memoryless).

Example 5 (Poisson and exponential agree). Calls arrive at $\lambda = 3$ per minute. Find P(no call in the next 30 seconds) in two ways.

Way 1 (Poisson): 30 seconds = 0.5 minute, so the count is Poisson(1.5). Way 2 (exponential): wait > 0.5 minute.

`fx` `=POISSON.DIST(0,1.5,FALSE)` → 0.2231

`fx` `=1-EXPON.DIST(0.5,3,TRUE)` → 0.2231

Answer: Both give 0.2231 = e^(−1.5). This is the link between the two distributions.

Example 6 (first failure of two machines). Two independent machines fail at rates 0.2 and 0.3 per hour (exponential lifetimes). Find the distribution of the time to the FIRST failure and P(first failure within 2 hours).

The minimum of independent exponentials is exponential and the rates add: $\lambda = 0.2 + 0.3 = 0.5$.

$$P(\min \le 2) = 1 - e^{-0.5 \times 2} = 1 - e^{-1} = 0.6321$$

`fx` `=EXPON.DIST(2,0.2+0.3,TRUE)` → 0.6321

Answer: Mean time to first failure = 1/0.5 = 2 hours. Probability 0.6321.

Example 7 (expected remaining wait). A bus wait is exponential with mean 4 minutes. You have already waited 6 minutes. What is the expected additional wait and the expected total wait?

Answer: By the memoryless property the additional wait is again exponential with mean 4 minutes. Expected total = 6 + 4 = 10 minutes. Waiting does not make the bus "due".

Example 8 (warranty and binomial link). A gadget lifetime is exponential with mean 5 years. Warranty = 1 year. Find P(claim). If 1000 gadgets are sold, find the expected number of claims and P(more than 200 claims).

$$p = P(X \le 1) = 1 - e^{-1/5} = 0.1813$$

Claims ~ Bin(1000, 0.1813), mean = 181.3, SD = 12.18.

fx `=1-BINOM.DIST(200,1000,0.1813,TRUE)` → 0.0586

Answer: p = 0.1813, expected claims 181.3, P(more than 200) = 0.0586.

5. Practice: 5 numericals

- λ = 0.1 per minute. Find P(X > 10).
- Mean waiting time 5 minutes. Find P(X ≤ 2).
- A machine part has mean life 200 hours. Find P(100 < X < 300).
- Mean wait 6 minutes. You have already waited 4 minutes. Find P(at least 3 more minutes).
- Mean 8 minutes. Find x such that P(X ≤ x) = 0.95.

Answer key (try on your own first, then check!)

1. =EXP(-0.1*10) → 0.3679
2. =EXPON.DIST(2,1/5,TRUE) → 0.3297
3. =EXPON.DIST(300,1/200,TRUE)-EXPON.DIST(100,1/200,TRUE) → 0.3834
4. memoryless: =EXP(-3/6) → 0.6065
5. =-LN(1-0.95)*8 → 23.9659

Code corner: Exponential

SciPy

```
from scipy.stats import expon
d = expon(scale=1/0.5)     # scale = 1/lambda = mean
d.sf(3)                    # P(X>3)
d.cdf(4) - d.cdf(1)        # P(1<X<4)
d.ppf(0.9)                 # 90th percentile
```

PyTorch

```
import torch
from torch.distributions import Exponential
d = Exponential(rate=torch.tensor(0.5))
1 - d.cdf(torch.tensor(3.))     # P(X>3)
d.icdf(torch.tensor(0.9))        # 90th percentile
```

TensorFlow Probability

```
import tensorflow_probability as tfp
d = tfp.distributions.Exponential(rate=0.5)
d.survival_function(3.)   # P(X>3)
d.quantile(0.9)
```

Normal Distribution

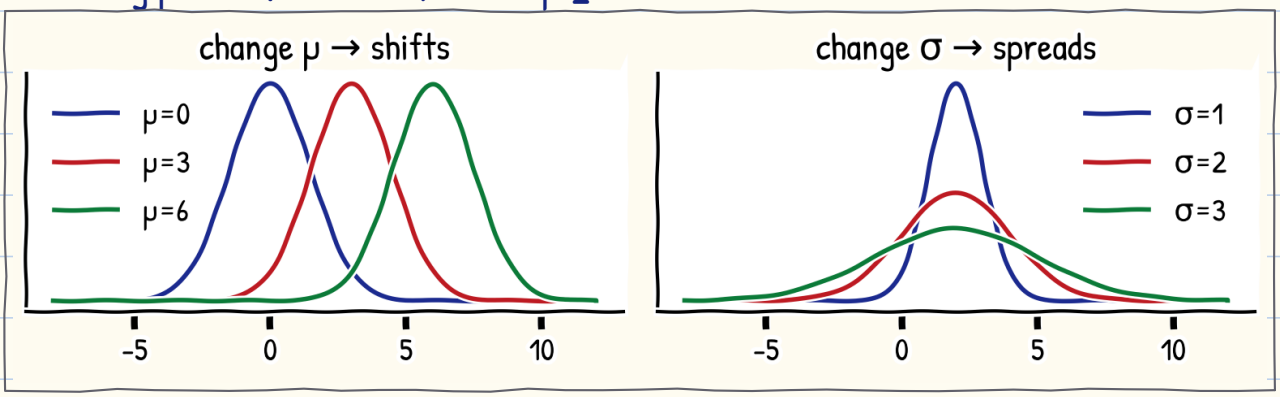
The bell curve: heights, errors, averages

1. Intuition

Heights, exam marks, measurement errors, packet weights: most values sit near the middle, and extreme values are rare on both sides. The result is a symmetric bell.

Why does it appear so often? A value that is the sum of many small independent pushes (genes, noise, tiny errors) tends to be bell-shaped. This is the Central Limit Theorem.

- μ (mean) decides WHERE the bell is centred.
- σ (standard deviation) decides HOW WIDE it is. Notation: $X \sim N(\mu, \sigma^2)$.
- Symmetric about μ: mean = median = mode. Tails never touch the axis.
- Turning points (inflection) are at $\mu \pm \sigma$.



Left: different μ. Right: different σ (same μ). Wider means flatter.

2. Formulas

PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

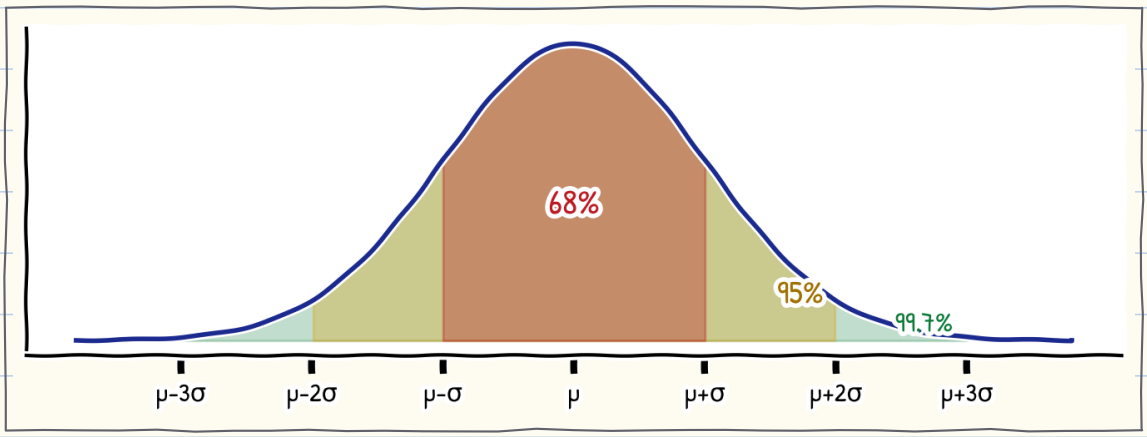
Reading the formula: the exponent $-(x-\mu)^2/(2\sigma^2)$ is 0 at $x = \mu$ (peak) and becomes more negative as x moves away, so the curve falls on both sides. The constant $1/(\sigma\sqrt{2\pi})$ makes the area exactly 1.

Standardising: the z-score

There is one table for every normal, because any $N(\mu, \sigma^2)$ can be converted to the standard normal $Z \sim N(0, 1)$. z tells you how many SDs x is from the mean.

z-score

$$z = \frac{x-\mu}{\sigma}, \qquad P(X \le x) = \Phi(z), \qquad \Phi(-z) = 1 - \Phi(z)$$



Empirical rule: 68% within 1σ, 95% within 2σ, 99.7% within 3σ.

Cumulative area Φ(z)	0.9	0.95	0.975	0.99	0.995
z value	1.282	1.645	1.960	2.326	2.576

3. Excel calculation

Quantity	Excel formula
$P(X \le x)$	=NORM.DIST(x, μ, σ, TRUE)
density f(x)	=NORM.DIST(x, μ, σ, FALSE)
$P(X > x)$	=1-NORM.DIST(x, μ, σ, TRUE)
$P(a < X < b)$	=NORM.DIST(b,μ,σ,TRUE)-NORM.DIST(a,μ,σ,TRUE)
z-score	=STANDARDIZE(x, μ, σ)
standard normal CDF	=NORM.S.DIST(z, TRUE)
x for a given area p	=NORM.INV(p, μ, σ)
z for a given area p	=NORM.S.INV(p)
simulation	=NORM.INV(RAND(), μ, σ)

Worksheet for $N(70, 8^2)$ (exam marks):

	A	B	C
	x	z	P(X≤x)
2	54	-2.00	0.0228
3	62	-1.00	0.1587
4	70	0.00	0.5000
5	78	1.00	0.8413
6	86	2.00	0.9772

fx B2: =STANDARDIZE(A2,70,8)

fx C2: =NORM.DIST(A2,70,8,TRUE)

4. Central limit theorem and approximations

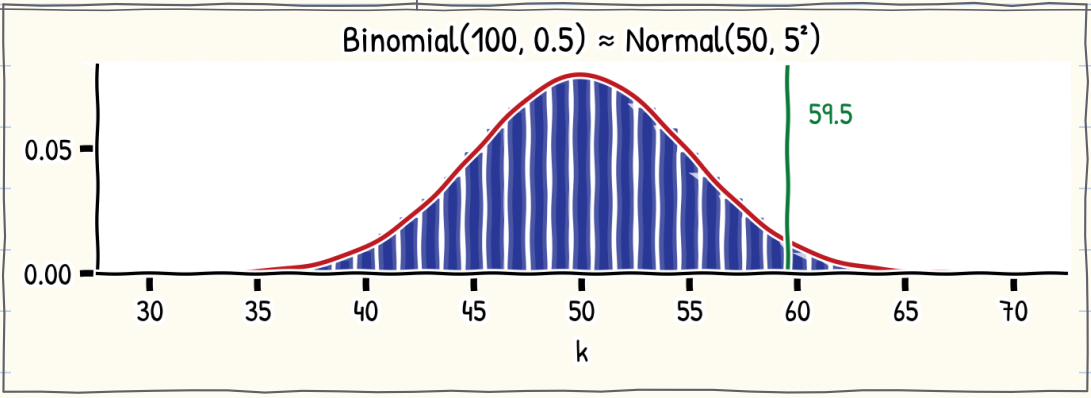
Averages of many observations are approximately normal, whatever the shape of the original data.

CLT

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \qquad z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Normal approximation to the Binomial: if $np \geq 5$ and $nq \geq 5$, then $Bin(n, p) \approx N(np, npq)$. Add or subtract 0.5 to move from the discrete bars to the continuous curve. This is the continuity correction.

Discrete wanted	Continuous used
$P(X \geq k)$	$P(Y > k - 0.5)$
$P(X \leq k)$	$P(Y < k + 0.5)$
$P(X = k)$	$P(k - 0.5 < Y < k + 0.5)$



The red curve hugs the blue bars. We use 59.5 as the cut-off for $P(X \geq 60)$.

5. Solved numericals

Example 1. Exam marks $X \sim N(70, 8^2)$. Find (a) $P(X > 80)$, (b) $P(60 < X < 75)$.

$$(a)\ z = \frac{80 - 70}{8} = 1.25, \quad P(X > 80) = 1 - \Phi(1.25) = 1 - 0.8944 = 0.1056$$

$$(b)\ z_1 = \frac{60 - 70}{8} = -1.25,\ z_2 = \frac{75 - 70}{8} = 0.625$$

$$P = \Phi(0.625) - \Phi(-1.25) = 0.7340 - 0.1056 = 0.6284$$

fx =NORM.DIST(75,70,8,TRUE)-NORM.DIST(60,70,8,TRUE) → 0.6284

Answer: (a) 0.1056 (b) 0.6284

Example 2. The top 10% of students get a distinction. Find the cut-off mark for $N(70, 8^2)$.

Need x with $P(X \le x) = 0.90$. From the table $z = 1.2816$.

$$x = \mu + z\sigma = 70 + 1.2816 \times 8 = 80.2524$$

`fx` `=NORM.INV(0.9,70,8)` → 80.2524

Example 3. Packet weight $X \sim N(500, 4^2)$ grams. Find $P(X < 492)$ and the percentage between 496 and 504 g.

$$z = \frac{492 - 500}{4} = -2 \Rightarrow P = 0.0228 \qquad 496 \text{ to } 504 = \mu \pm 1\sigma \Rightarrow 0.6827$$

Answer: 0.0228 (about 2.3%) and 68.27%

Example 4 (approximation). A fair coin is tossed 100 times. Find P(at least 60 heads) with the normal approximation.

$\mu = np = 50$, $\sigma = \sqrt{(100 \cdot 0.5 \cdot 0.5)} = 5$. Continuity correction: $P(X \ge 60) \to P(Y > 59.5)$.

$$z = \frac{59.5 - 50}{5} = 1.9 \Rightarrow P = 1 - \Phi(1.9) = 0.0287$$

`fx` `=1-BINOM.DIST(59,100,0.5,TRUE)` → 0.0284 (exact)

Answer: Approximation 0.0287 vs exact 0.0284. Very close.

Example 5 (CLT). Individual values have mean 50 and SD 10. For a sample of n = 25 find P(sample mean > 53).

$$\sigma_{\bar{x}} = \frac{10}{\sqrt{25}} = 2, \quad z = \frac{53 - 50}{2} = 1.5, \quad P = 1 - \Phi(1.5) = 0.0668$$

`fx` `=1-NORM.DIST(53,50,10/SQRT(25),TRUE)` → 0.0668

5b. More solved numericals: going deeper

Example 6 (find μ and σ from two facts). A test score X is normal with $P(X < 45) = 0.1587$ and $P(X < 65) = 0.9772$. Find μ and σ.

From the table $\Phi(-1) = 0.1587$ and $\Phi(2) = 0.9772$, so 45 is 1 SD below the mean and 65 is 2 SD above.

$$\mu - \sigma = 45, \quad \mu + 2\sigma = 65 \Rightarrow 3\sigma = 20, \ \sigma = 6.667, \ \mu = 51.667$$

fx `=NORM.S.INV(0.1587)` → -1.000 (≈ −1)

Example 7 (sum of independent normals). Task A time ~ N(30, 4²) min and task B time ~ N(45, 6²) min, independent. Find P(total time > 90).

Means add. Variances add (not SDs): Var = 16 + 36 = 52, SD = √52 = 7.2111.

$$T \sim N(75, 52), \qquad z = \frac{90 - 75}{7.2111} = 2.080, \quad P = 0.0188$$

fx `=1-NORM.DIST(90,75,SQRT(4^2+6^2),TRUE)` → 0.0188

Careful! SD(A + B) is NOT 4 + 6 = 10. Add the VARIANCES, then take the square root.

Example 8 (finance: one-day Value at Risk). Daily return ~ N(0.05%, 1.2%²). Find the 95% one-day VaR for a ₹10,00,000 portfolio.

VaR uses the 5th percentile of returns: z = −1.645.

$$r_{5\%} = 0.05 + (-1.645)(1.2) = -1.9238\%$$

fx `=NORM.INV(0.05,0.05,1.2)` → -1.9238

Answer: With 95% confidence the one-day loss will not exceed 1.924% = ₹19,238.

Example 9 (process capability). Bolt diameter ~ N(10, 0.05²) mm. Specification: 9.90 to 10.10. Find the fraction within spec, defects per 10,000, and the σ needed for 99% within spec.

$$z = \pm\frac{0.10}{0.05} = \pm 2 \Rightarrow P = 0.9545$$

fx `=NORM.DIST(10.1,10,0.05,TRUE)-NORM.DIST(9.9,10,0.05,TRUE)` → 0.9545

Defects = 4.55% = about 455 per 10,000.

For 99% within ±0.10 we need z = NORM.S.INV(0.995) = 2.5758, so σ = 0.10/2.5758.

Answer: 0.9545 in spec, about 455 defects per 10,000; required σ = 0.0388 mm.

Example 10 (P(X = k) with continuity correction). X ~ Bin(50, 0.5). Approximate P(X = 25).

$$P(24.5 < Y < 25.5) = \Phi\left(\frac{0.5}{3.5355}\right) - \Phi\left(\frac{-0.5}{3.5355}\right) = 0.1125$$

fx	=BINOM.DIST(25,50,0.5,FALSE)	→ 0.1123 (exact)

Answer: Approximation 0.1125 vs exact 0.1123. A single bar is replaced by a width-1 slice of the curve.

6. Practice: 5 numericals

- Heights $X \sim N(170, 6^2)$ cm. Find $P(X > 180)$.
- Salaries $X \sim N(50000, 8000^2)$. Find $P(40000 < X < 60000)$.
- Battery life $X \sim N(24, 2.5^2)$ hours. Find $P(X < 20)$.
- IQ $\sim N(100, 15^2)$. Find the 95th percentile.
- $X \sim \text{Bin}(80, 0.4)$. Use the normal approximation (with continuity correction) for $P(X \leq 28)$ and compare with the exact value.

Answer key (try on your own first, then check!)

1. =1-NORM.DIST(180,170,6,TRUE) → 0.0478
2. =NORM.DIST(60000,50000,8000,TRUE)-NORM.DIST(40000,50000,8000,TRUE) → 0.7887
3. =NORM.DIST(20,24,2.5,TRUE) → 0.0548
4. =NORM.INV(0.95,100,15) → 124.6728
5. $\mu = 32$, $\sigma = 4.3818$: approx 0.2122; exact =BINOM.DIST(28,80,0.4,TRUE) → 0.2131

Code corner: Normal

SciPy

```
from scipy.stats import norm
d = norm(loc=70, scale=8)
d.sf(80)                  # P(X>80)
d.cdf(75) - d.cdf(60)     # P(60<X<75)
d.ppf(0.90)               # cut-off for top 10%
```

PyTorch

```
import torch
from torch.distributions import Normal
d = Normal(loc=70., scale=8.)
1 - d.cdf(torch.tensor(80.))   # P(X>80)
d.icdf(torch.tensor(0.90))     # 90th percentile
```

TensorFlow Probability

```
import tensorflow_probability as tfp
d = tfp.distributions.Normal(loc=70., scale=8.)
d.survival_function(80.)   # P(X>80)
d.quantile(0.90)
```

Mixed Practice

First decide WHICH distribution, then calculate

In exams the hard part is choosing the model. For each problem, write the distribution with its parameters, then use Excel.

- 10 emails arrive; each is spam with probability 0.3, independently. Find P(exactly 4 are spam).
- A website gets 12 visits per minute on average. Find P(more than 15 visits in a minute).
- The gap between visits is exponential at 12 visits per minute (0.2 per second). Find P(next visit within 3 seconds).
- A courier arrives uniformly between 2:00 pm and 2:40 pm. Find P(arrival before 2:10 pm).
- A stock's daily return is $N(0.1\%, 1.5\%^2)$. Find P(the stock makes a loss tomorrow).
- 3% of loans in a portfolio default. In 200 loans find P(at most 8 defaults), exactly and with the Poisson approximation.
- Component life is exponential with mean 500 hours. Three independent components are used. Find P(all three last beyond 300 hours).
- Invoices have mean 1000 and SD 120. For 36 invoices find P(sample mean < 970).

Answer key (try on your own first, then check!)

1. Bin(10, 0.3): =BINOM.DIST(4,10,0.3,FALSE) → 0.2001
2. Poisson(12): =1-POISSON.DIST(15,12,TRUE) → 0.1556
3. Exp(0.2 per s): =EXPON.DIST(3,0.2,TRUE) → 0.4512
4. U(0, 40) minutes: =(10-0)/(40-0) → 0.2500
5. $N(0.1, 1.5^2)$: =NORM.DIST(0,0.1,1.5,TRUE) → 0.4734
6. exact =BINOM.DIST(8,200,0.03,TRUE) → 0.8504; Poisson(6) =POISSON.DIST(8,6,TRUE) → 0.8472
7. each: =EXP(-300/500) → 0.5488; all three: =EXP(-300/500)^3 → 0.1653
8. SE = 120/√36 = 20: =NORM.DIST(970,1000,20,TRUE) → 0.0668

Hint: For problem 7: independent events multiply. P(all three) = $(0.5488)^3$ = e^(−1.8).

Connections & Excel Map

How the five distributions fit together

- Bernoulli → Binomial: add up n independent yes/no trials.
- Binomial → Poisson: n large, p small, $\lambda = np$ (rule of thumb $n \geq 20$, $p \leq 0.05$).
- Binomial → Normal: $np \geq 5$ and $nq \geq 5$, use $N(np, npq)$ with the ± 0.5 continuity correction.
- Poisson ↔ Exponential: counts per interval are Poisson(λt); gaps between events are Exp(λ), with mean gap $1/\lambda$.
- Poisson → Normal: for large λ (about 10 or more) the bars look like a bell.
- CLT: averages of many independent values are normal, whatever the original shape.

Uniform is the seed of every simulation. If $U \sim U(0, 1)$, then any distribution can be generated by $X = F^{-1}(U)$. For the exponential this is a one-liner (the Excel simulation formula above):

inverse CDF

$$X = \frac{-\ln U}{\lambda} \sim \text{Exp}(\lambda), \qquad U \sim U(0, 1)$$

Excel function map

Distribution	Single value / cumulative	Inverse and simulation
Binomial	BINOM.DIST(k,n,p,FALSE) BINOM.DIST(k,n,p,TRUE)	BINOM.INV(n,p,α) BINOM.INV(n,p,RAND())
Poisson	POISSON.DIST(k,λ,FALSE) POISSON.DIST(k,λ,TRUE)	no inverse: scan the cumulative column
Uniform	=1/(b-a) =(x-a)/(b-a)	=a+p*(b-a) =a+(b-a)*RAND()
Exponential	EXPON.DIST(x,λ,FALSE) EXPON.DIST(x,λ,TRUE)	=-LN(1-p)/λ =-LN(RAND())/λ
Normal	NORM.DIST(x,μ,σ,FALSE) NORM.DIST(x,μ,σ,TRUE)	NORM.INV(p,μ,σ) NORM.INV(RAND(),μ,σ)

Cheat Sheet

One page to revise before the exam

Dist.	Mass / density	Mean	Var	Excel
Binomial Bin(n,p)	C(n,k) p^k q^(n−k)	np	npq	BINOM.DIST(k,n,p,FALSE/TRUE)
Poisson Poi(λ)	e^(−λ) λ^k / k!	λ	λ	POISSON.DIST(k,λ,FALSE/TRUE)
Uniform U(a,b)	1/(b−a)	(a+b)/2	(b−a)²/12	(x−a)/(b−a) [by hand]
Exponential Exp(λ)	λ e^(−λx), P(X>x)=e^(−λx)	1/λ	1/λ²	EXPON.DIST(x,λ,TRUE)
Normal N(μ,σ²)	bell, z = (x−μ)/σ	μ	σ²	NORM.DIST(x,μ,σ,TRUE)

Which distribution?

- Count of successes in a FIXED number of yes/no trials → Binomial.
- Count of events in an interval, no upper limit, given an average rate → Poisson.
- Time until the NEXT event (rate λ known) → Exponential.
- Anything equally likely across a range → Uniform.
- Bell-shaped measurements, or averages / sums of many values → Normal.

Exam checklist and common mistakes

- ✔ Write the distribution and its parameters first: X ~ Bin(10, 0.5).
- ✔ P(X ≥ k) = 1 − P(X ≤ k−1) for discrete RVs. For continuous RVs the ≥ vs > does not matter.
- ✔ Poisson: always adjust λ to the length of the interval (λ = rate × time).
- ✔ Exponential: Excel uses the RATE λ, not the mean. Mean = 1/λ.
- ✔ Normal: standardise, draw a quick sketch, shade the area, then use NORM.DIST.
- ✔ Uniform: probability = length ÷ (b − a). Check that your answer is between 0 and 1.
- ✔ Round only at the end and quote 4 decimal places.

Final check: Probability answers must always lie between 0 and 1. If you get something outside this range, re-check the formula.